Practice workbook · State or explain safety precautions for a practical

Burning food, hot or cold water, reagents, glass, mouthpieces, spirometers and handling cells or bacteria: hazards and how to reduce risk.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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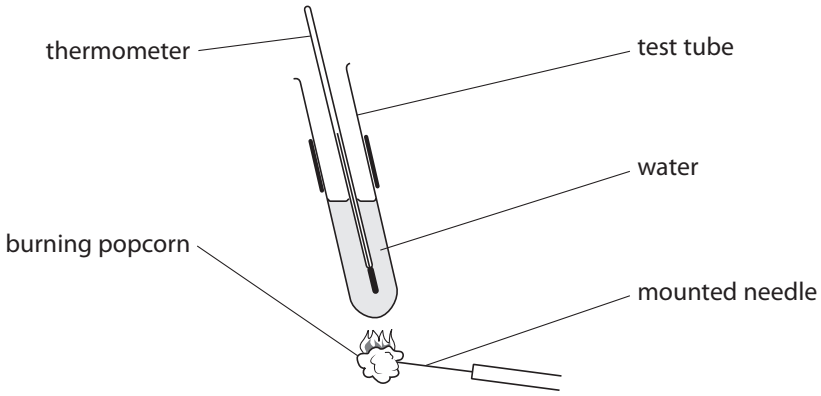
Type ID: HBI-PRAC-SAFETY-HAZARD-PRECAUTION

1. Question 2 · part a.ii · 2 marks

November 2020 Paper 1

Worked-example retry · Attempt part a.ii only.

2 A student uses this apparatus to investigate the energy contained in popcorn.



This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

(1)

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(ii) Explain one safety precaution that the student should take during the investigation.

(2)

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1. Question 2 · part a.ii · 2 marks

November 2020 Paper 1

Worked-example retry · Attempt part a.ii only.

(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

(3)

energy released = ... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

(3)

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1. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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2. Question 3 · part d · 2 marks

June 2019 Paper 2

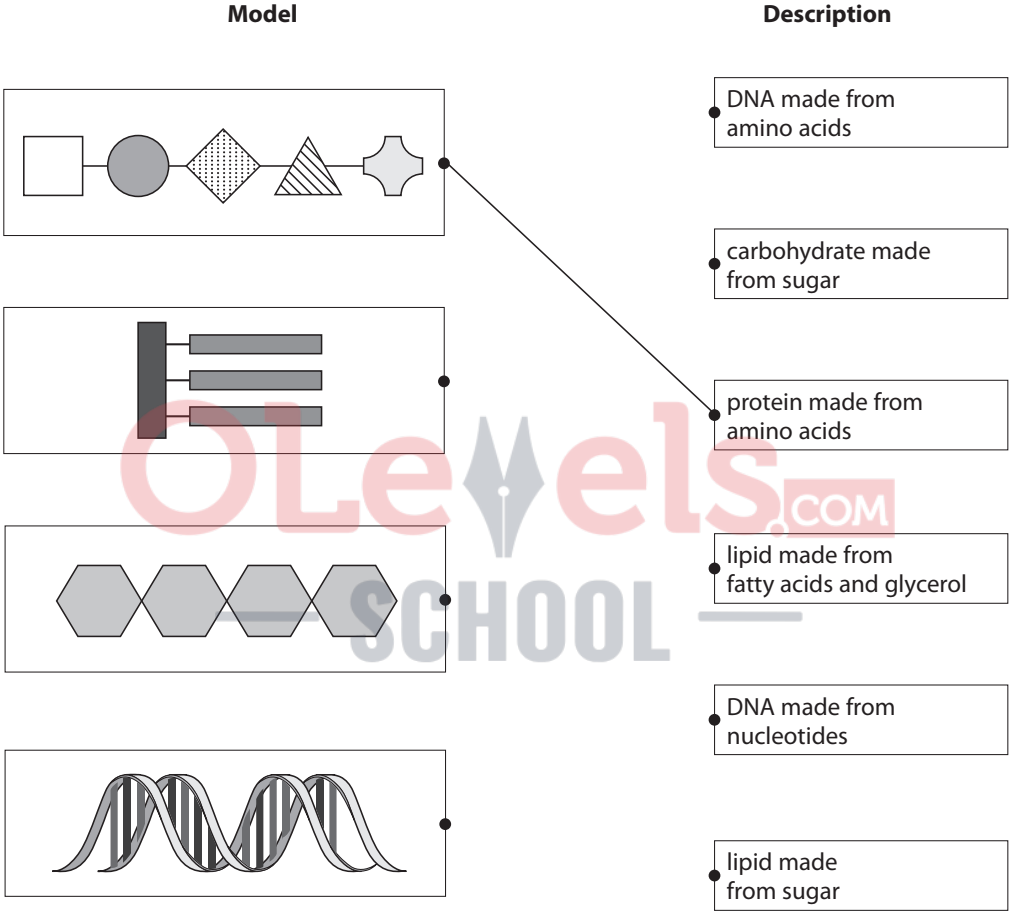
Further practice · Attempt part d only.

3 (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

(3)



(b) Which body organ produces enzymes that break down protein?

(1)

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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2. Question 3 · part d · 2 marks

June 2019 Paper 2

Further practice · Attempt part d only.

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(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

Substance	Colour after Biuret test
milk	purple
pasta	blue
lemon juice	blue
cheese	purple
distilled water	purple

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(i) In the student's test, how many of the substances give a positive result for protein?

(1)

.....

(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

(1)

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.....

(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

(2)

Hazard	Reducing risk
stools – trip hazard	keep stools under bench
broken glass – cuts	
Biuret reagent – irritant	

(Total for Question 3 = 8 marks)

2. Question 3 · part d · 2 marks · Working and answer

Working and answer

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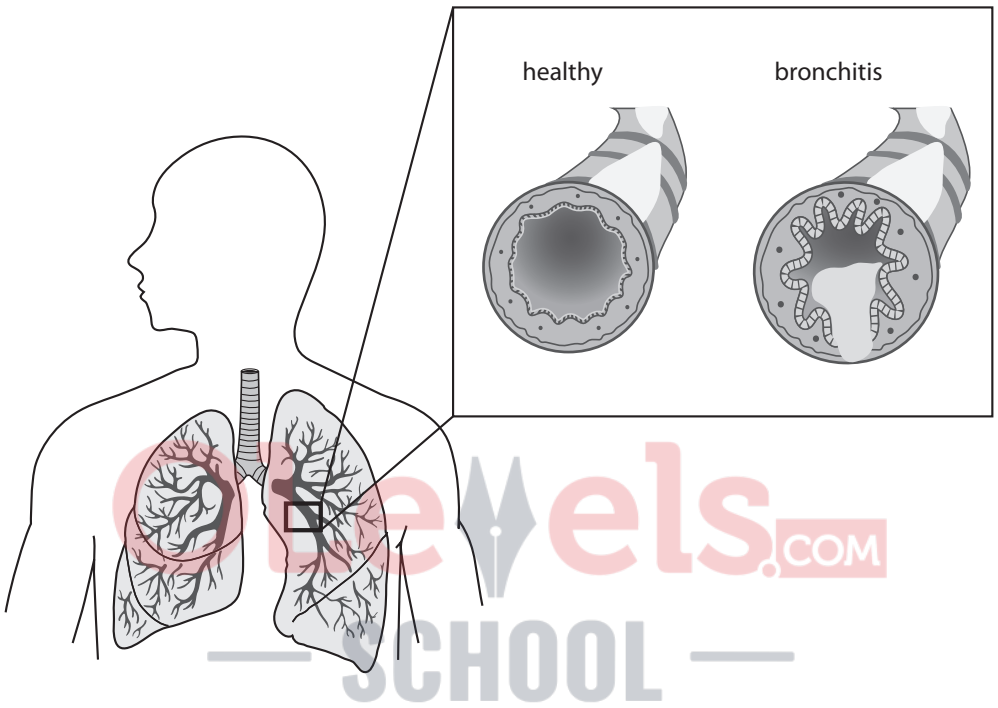
3. Question 6 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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3. Question 6 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

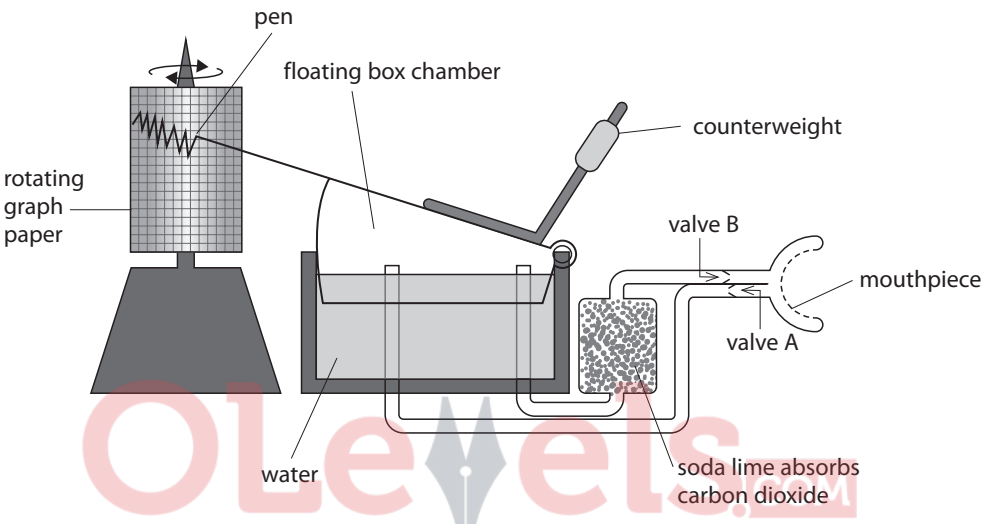
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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3. Question 6 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

(ii) State one way to reduce risk to health when using a spirometer.

(1)

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(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

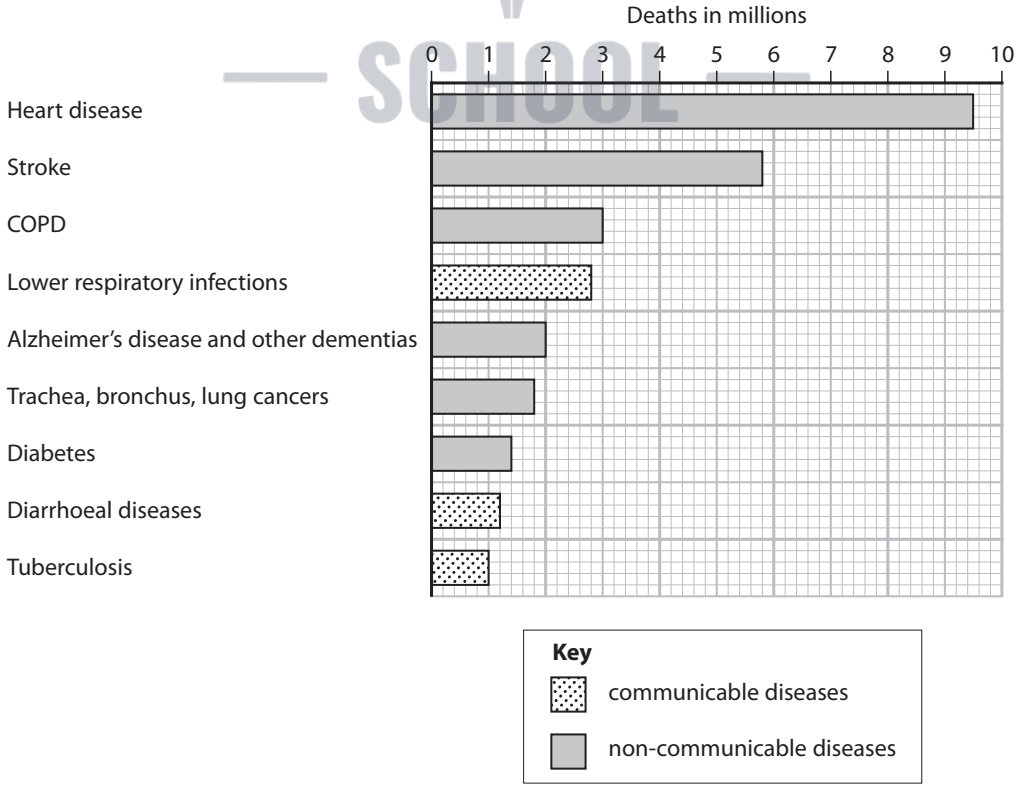
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(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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3. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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4. Question 3 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

3 (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

(2)

	Steps
	select a suitable objective lens
5	adjust focusing wheel to obtain a clear image
1	remove cells from the inside of the cheek
	place microscope slide onto the microscope stage
	smear the cell sample onto a microscope slide

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

(1)

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.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

(1)

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(iv) Give two ways that the student could avoid health hazards when working with living cells.

(2)

1 .....

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2 .....

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4. Question 3 · part a.iv · 2 marks

November 2020 Paper 1R

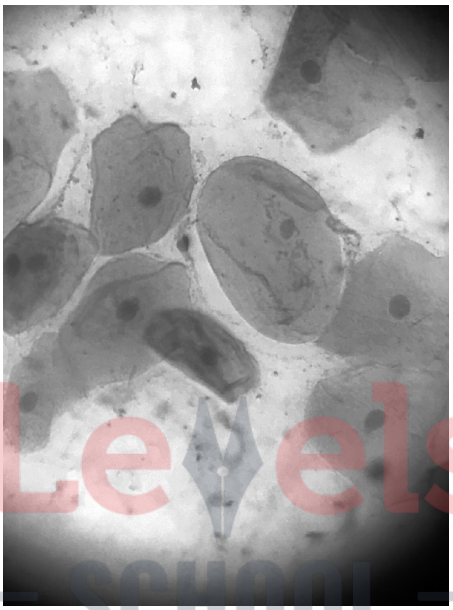
Further practice · Attempt part a.iv only.

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(b) The photograph shows an image of a cheek cell seen under a light microscope using a 10× eyepiece lens and a 40× objective lens.



© Esmiles/Shutterstock

(i) Calculate the total magnification used to view the cheek cell.

(1)

total magnification = ...

(ii) The actual diameter of the cheek cell shown in the photograph is 0.018 mm.

Convert the diameter of the cheek cell into micrometre (μm).
[1 μm = 10^{-6} m]

(2)

diameter = ... μm

(Total for Question 3 = 9 marks)

4. Question 3 · part a.iv · 2 marks · Working and answer

Working and answer

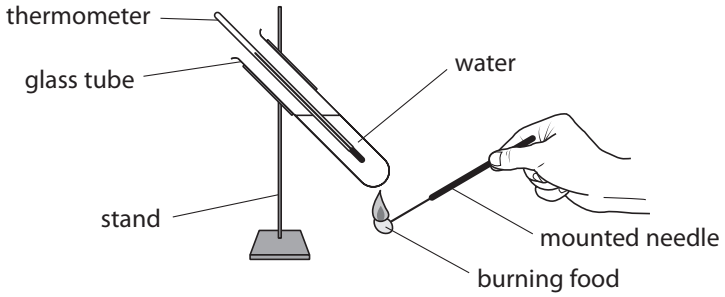
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5. Question 7 · part a.i · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

7 A student uses this apparatus to investigate the energy content of different foods.



This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

(3)

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2 .....

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3 .....

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P 6 6 4 3 2 A 0 1 8 2 4

5. Question 7 · part a.i · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

(ii) State three factors that should be controlled during the investigation.

(3)

1 .....

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2 .....

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3 .....

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(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

(5)

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5. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

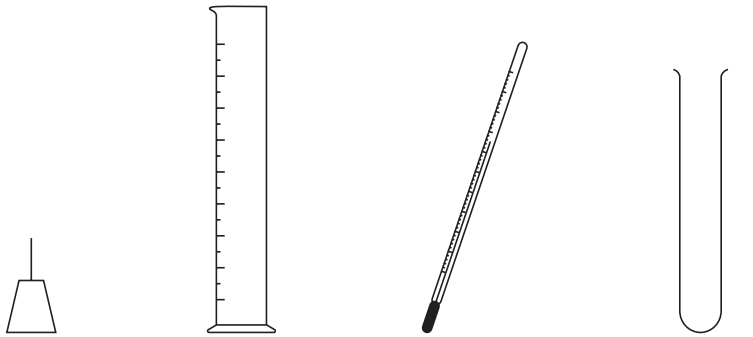
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6. Question 3 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

3 (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

(5)

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6. Question 3 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(ii) Name two variables that need to be controlled in the investigation.

(2)

1 .....

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2 .....

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(iii) Give two safety precautions needed for this investigation.

(2)

1 .....

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2 .....

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6. Question 3 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

Daily Energy Requirement in kcal

Age of adults in years	Males	Females
19–24	2772	2175
25–34	2749	2175
35–44	2629	2103
45–54	2581	2103
55–64	2581	2079
65–74	2342	1912
75 and over	2294	1840

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

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(i) Describe the patterns in the energy requirements for males and for females.

(2)

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

(1)

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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6. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

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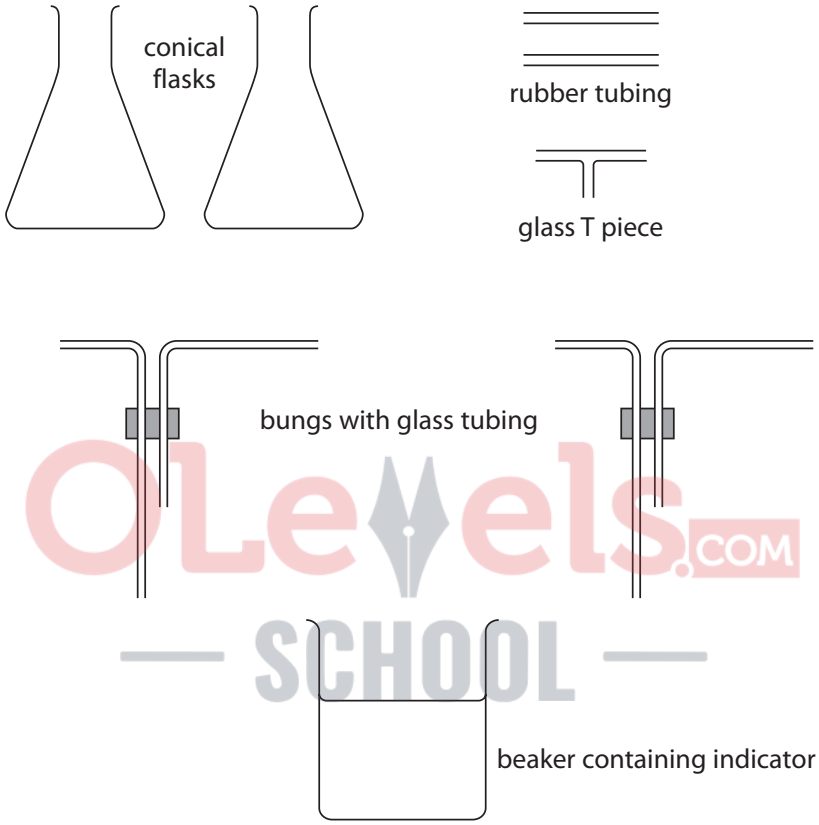
7. Question 1 · part d · 1 mark

June 2022 Paper 2R

Further practice · Attempt part d only.

1 A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.



The apparatus is not drawn to scale.

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7. Question 1 · part d · 1 mark

June 2022 Paper 2R

Further practice · Attempt part d only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

(3)

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(b) Name an indicator that could be used in the investigation.

(1)

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

(4)

7. Question 1 · part d · 1 mark

June 2022 Paper 2R

Further practice · Attempt part d only.

(d) State one safety precaution.

(1)

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(e) State one variable that should be controlled.

(1)

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(Total for Question 1 = 10 marks)

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Working and answer

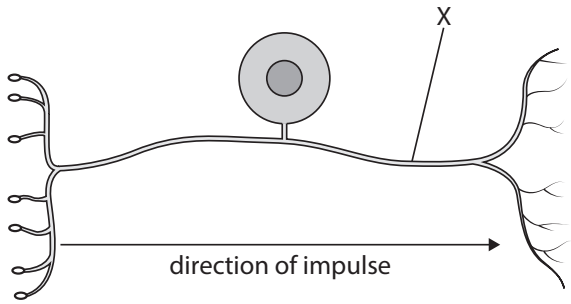
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8. Question 1 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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8. Question 1 · part c.i · 1 mark

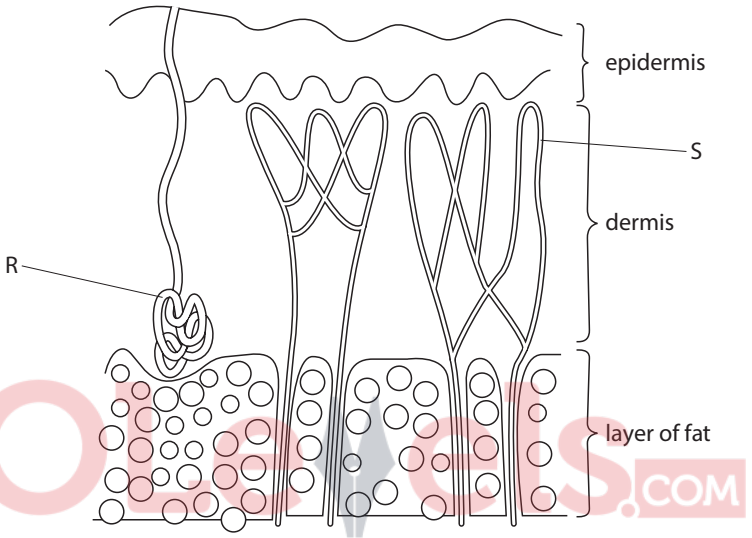
June 2024 Paper 1

Further practice · Attempt part c.i only.

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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

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.....

8. Question 1 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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8. Question 1 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

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(c) A student investigates the effect of temperature on receptors in the skin.

This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

(1)

.....

.....

(ii) Describe how to reduce the risk of this hazard.

(2)

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(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

(1)

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(Total for Question 1 = 12 marks)

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8. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

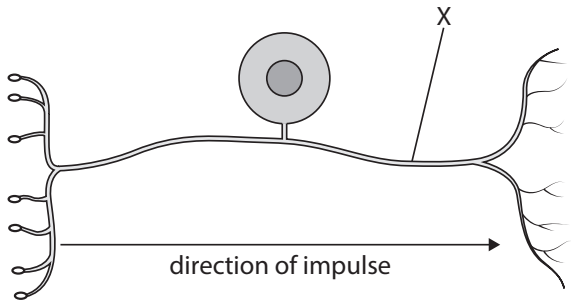
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9. Question 1 · part c.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part c.ii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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9. Question 1 · part c.ii · 2 marks

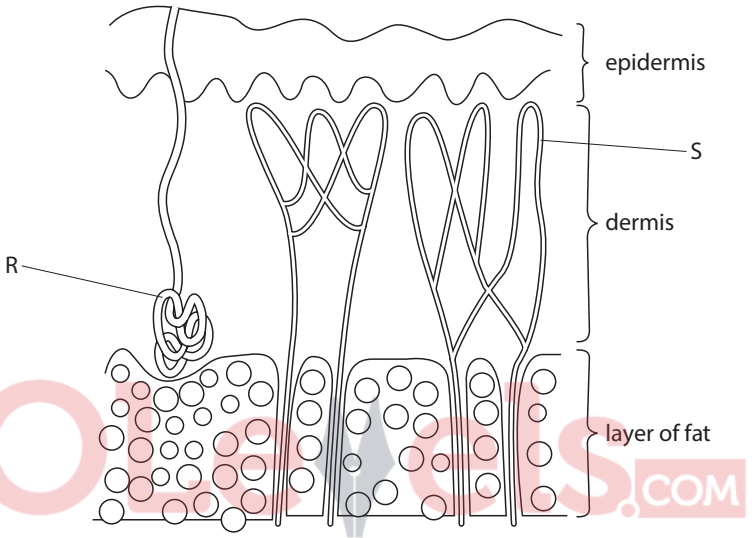
June 2024 Paper 1

Further practice · Attempt part c.ii only.

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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

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9. Question 1 · part c.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part c.ii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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9. Question 1 · part c.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part c.ii only.

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This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

(1)

.....

.....

(ii) Describe how to reduce the risk of this hazard.

(2)

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.....

(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

(1)

.....

.....

(Total for Question 1 = 12 marks)

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9. Question 1 · part c.ii · 2 marks · Working and answer

Working and answer

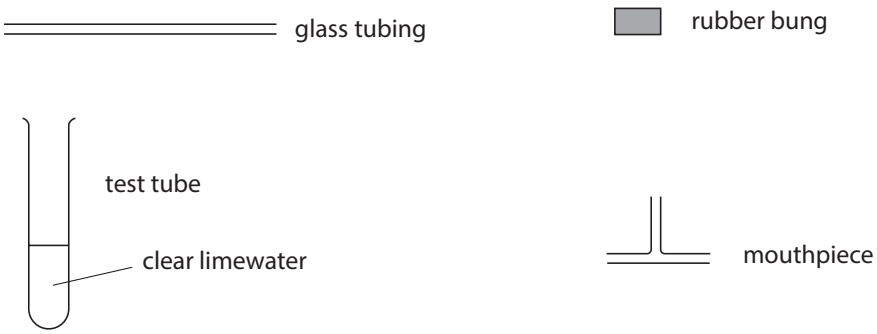
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10. Question 4 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

4 (a) The diagram shows some scientific equipment.



(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

(3)

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10. Question 4 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

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(ii) State one safety precaution in this investigation.

(1)

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

Mean breathing rate in breaths per minute		
at rest	**after gentle exercise**	**after vigorous exercise**
14	19	50

(i) Give one control variable in this investigation.

(1)

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

(4)

(Total for Question 4 = 9 marks)

10. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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11. Question 8 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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11. Question 8 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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11. Question 8 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

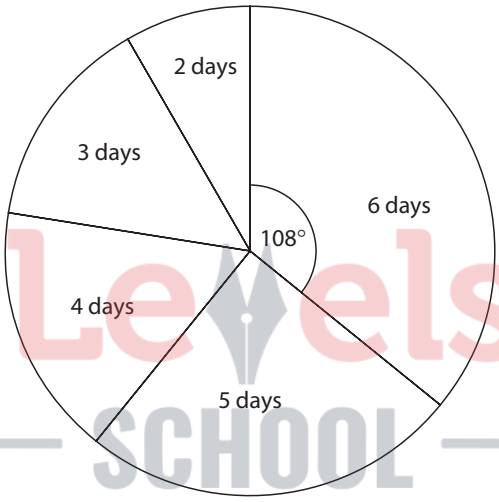
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(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.



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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

(3)

number of types of antibiotic = ...

P 7 4 6 3 2 A 0 2 3 2 4

Turn over ▶

11. Question 8 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

(ii) State a reason why it is important to wear gloves and a face mask when investigating bacteria.

(1)

(iii) Describe how the use of antibiotics can cause antibiotic-resistant strains of bacteria.

(2)

(Total for Question 8 = 15 marks)

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Working and answer